

FOURTH ANNUAL IN-HOUSE CONVENTION

DEPARTMENT OF PSYCHOLOGY

DALHOUSIE UNIVERSITY

APRIL 26-27, 1978

34

SCHEDULE

April 26

April 27

9:00 - 9:30 COFFEE

9:30 - 9:45
9:45 - 10:00
10:00 - 10:15
10:15 - 10:30

COFFEE - 10:30 - 11:00

11:00 - 11:15
11:15 - 11:30
11:30 - 11:45
11:45 - 12:00

LUNCH - 12:00 - 1:30

1:30 - 1:45
1:45 - 2:00
2:00 - 2:15
2:15 - 2:30

COFFEE - 2:30 - 3:00

3:00 - 3:15
3:15 - 3:30
3:30 - 3:45
3:45 - 4:00
4:00 - 4:15

Ch. Regan

1. M. Yoon
2. R. Brown
3. J. Stevenson
4. L. Davies

Ch. Sutherland

5. Beverley & Regan
6. B. McNaughton
7. L. White
8. T. Whalen

Ch. Yoon

9. Tansley & Regan
10. R. Douglas
11. J. Gardner
12. Regan & Beverley

Ch. Willner

13. Tower & Barresi
14. Baxter & Barresi
15. P. Nau
16. C. Barnes
17. Day & MacKay

Ch. Whalen

18. J. Fentress
19. Z. Havkin
20. M. Cynader
21. S. Nakajima

Ch. Havkin

22. Allen & Barresi
23. B. Rusak
24. A. Randich
25. F. Baker

Ch. Rusak

26. M. Ozier
27. K. Bloom
28. T. Whalen
29. Barresi & Dalzell

Ch. Barresi

30. J. Willner
31. J. Raymond
32. Klein, Day, & Armitage
33. N. Bennett
34. R. Sutherland

ABSTRACTS

1. M. Yoon - Neuritic Outgrowths from the Adult Goldfish Retinal Tissue in Culture.

The patterns of neuritic outgrowths from the retina are investigated in vitro under various experimental conditions by culturing the retinal tissue, explanted from the adult goldfish. Under favorable culture conditions, the retinal tissue shows vigorous outgrowth of neurites as early as 12 hr and as late as 4-5 weeks after explantation. The neuritic outgrowths from the retinal explant is enhanced by section of the axons of the retinal ganglionic cells prior to the explantation of the retinal tissue. The direction of neuritic outgrowths from the retinal explant in vitro seems to follow the regularity in the embryonic development of neuronal processes in vivo: the retinal neurites tend to grow in the radial direction towards the optic disc at the center of the retina, especially at an early stage in culture. In further experiments, a strip of retinal tissue and its topographically matching area of the optic tectum are co-cultured in various geometric configuration within the same culture chamber. Some of the retinal neurites eventually grow into the tectal tissue. It is not known as yet whether these retinal neurites re-establish functional synapses with appropriate visual neurones of the tectal tissue in vitro.

2. R. Brown - Measuring Copulatory Rate: A Mount-Bout Analysis.

The male rat's rate of copulation (intromissions per minute) has been used as a reflection of his level of sexual arousal. I have shown that copulatory rate is not a primary measure, but is the product of two more basic measures: mount-bout rate and the probability of obtaining an intromission on each mount-bout. Mount-bout rate is shown to be a reflection of arousal level while the probability of obtaining an intromission reflects the male's performance mechanism and the level of cooperation shown by the female. The use of mount-bouts is necessary for the dissociation of arousal and performance mechanisms controlling male rat sexual behaviour. *How often do they come up?*

3. J. Stevenson - Growth of the Goldfish Optic Tectum.

Goldfish grow throughout their lives; their brains grow also. It is apparent that this growth occurs through addition of new cells and by increases in size of existing cells. I will describe the post-natal growth of the tectal stratum periventriculare and show that most of the growth in this region occurs through enlargement of existing cells. In light of growth by cell addition in the retina, this form of tectal growth is difficult to understand. The occurrence of continuously shifting synaptic connections, and the implications of this phenomenon in an active, functioning animal, will be discussed. *FOR AIR?*

4. L. Davies - A Glimpse at Scaling Theory.

This is not another talk about fish, but rather a brief introduction to a mathematical approach which can lead to insights about such problems as the general function of a brain and the inner experience of animals.

5. K. Beverley & D. Regan - Changing-size Channels for Different Object Sizes.

We previously reported psychophysical evidence for the existence of a neural mechanism specifically sensitive to changing object size. We now report that for narrow adapting stimuli, the changing-size specific threshold elevation is significant, providing the widths of the test and adapting stimuli do not differ by more than 0.5° . However, no significant elevation occurs for adapting stimuli wider than 1.5° . Further evidence that the changing-size channel is sensitive only to objects smaller than 1.5° is that aftereffects to changing size are explicable in terms of Waterfall effects at the moving edges when adapting stimuli are wider than 1.5° .

6. B. McNaughton - Comparison of Synaptic Modifiability in the Medial and Lateral Perforant Pathways: Evidence for two Functionally Discrete Subsystems.

The synapses of the perforant pathway from the entorhinal cortex to the fascia dentata are organized in a clearly laminar fashion. Those fibres originating most laterally in the entorhinal area terminate most distally on the granule cell dendrites, while the most medially originating fibres terminate most proximally. This system has been classically regarded as a functional unit. Physiological studies of short-term synaptic plasticity, however, suggest two functionally discrete subsystems. This data is in agreement with recent histochemical studies.

7. L. White - Social Play Levels in Young Rhesus Monkeys.

Levels of social play of immature rhesus monkeys living in small captive groups are extremely variable, and depend, to some extent, on the composition of the social group. It can be difficult to pick out developmental changes in the social play tendencies of a single individual, as any change in interaction level or type is the result of changes in the behaviour of both interacting animals. Here the importance of studying individual behaviour in the context of the social group is emphasized, and an attempt is made to assign responsibility for changes in play level more to one play partner than to the other.

8. T. Whalen - Remembering Processes in the Pigeon.

Depending on the requirements of the situation, it is believed that an organism, such as a pigeon, can either remember a stimulus or a future behaviour required by the present stimulus. Honig has shown that in situations in which a pigeon is only required to remember whether to peck or not, remembering occurs over longer intervals than are normally found with procedures which require the recall of a stimulus. The present research uses a procedure which also requires only that the pigeon remember whether to respond or not. It is simpler than those that have been used previously. Since this seems to produce as much recall in pigeons as Honig's procedure, this hypothesis is supported.

is color important?

9. B. Tansley & M. Regan - Estimation of the Sensitivity of FM Channels in the Human Auditory System.

Because of the high sensitivity of auditory mechanisms to amplitude changes associated with frequency variations of an acoustic stimulus, the estimation of frequency-modulation sensitivity via a threshold method is often contaminated with input from channels selectively sensitive to changing amplitude. We propose a method whereby it appears possible to estimate the sensitivity of FM channels by "titrating off" the activity of AM channels due to differential responses within the auditory system to equal amplitude frequency variations. With this method, the sensitivity of the FM channel under test appears to be lower than previously reported.

10. R. Douglas - Heterosynaptic Inhibition of Synaptic Plasticity.

There are two major inputs to the dentate area of the hippocampus. The perforant path fibers make excitatory synapses onto the granule cells, and long-lasting increases in the effectiveness of these synapses can be produced by high-frequency activation. The commissural input has a predominately inhibitory effect on the granule cells, and does not appear to be modifiable. However, high-frequency activity along it can completely prevent the enhancement of the perforant path synapses. The commissural effect on synaptic plasticity can be dissociated from its effects on cell discharge.

11. J. Gardner - Induced Anisocoria.

Kittens raised with a dimming filter over one eye show long-lasting changes in the pupillary light reflex. The deprived eye becomes more sensitive to light, and when the filter is removed, that pupil is consistently smaller. The anisocoria lasts for weeks. We suspect that this plasticity is due to a cerebellar mechanism and are now testing normal and lesioned adult cats.

12. D. Regan & K. Beverley - Resolution of Ambiguity in the Perception of Motion in Depth.

Increasing or decreasing retinal image size is an ambiguous visual cue because it can mean either that an object's size is changing or that the object is moving in depth. How does the visual system resolve this ambiguity so as to provide an unequivocal basis for action? We will present evidence that there are separate information-processing channels for changing size and for motion in depth, and that the visual system contains a biasing mechanism that is capable of suppressing the activity of one or other channel. One trigger that activates this biasing mechanism is the relationship between the speeds of the horizontal and vertical edges: if the object's shape is constant, then the changing-size channel is rendered less effective.

13. G. Tower & J. Barresi - The Effects of a Third Choice on Cooperation in a Mixed-Motive Game.

The generally low amount of cooperative (30%) as opposed to competitive (70%) responding in the two-choice, prisoner's dilemma game may be due to the meanings attached to the restricted choices provided by the game. In the present study a third choice was added whose mere presence changed the "meaning" of the cooperative and competitive choices. Three "meanings" of the cooperative choice were compared - sharing, self-interest, and altruism. The sharing condition produced the most cooperative responding (sharing = 56%, self-interest = 33%, altruism = 36%). The self-interest condition produced the most competitive responding (sharing = 25%, self-interest = 43%, altruism = 35%). Finally, the altruism condition produced the most third (contract) choice responding (sharing = 19%, self-interest = 24%, altruism = 29%).

14. R. Baxter & J. Barresi - The Effects of Three Preprogrammed Strategies on Cooperative responding in a Three-Choice Mixed-Motive Game.

Cooperative responding in the prisoner's dilemma game has been increased from a free play 30% to 80% when a preprogrammed tit-for-tat strategy is used by the "partner". This strategy involves responding on each trial with the real subject's previous response. Thus, the subject is rewarded for a cooperative response (i.e., with cooperation) and punished for a competitive response (i.e., with competition). The effectiveness of this and two other preprogrammed strategies were compared in a three-choice mixed-motive game. The results of the experiment suggest that the reward and punish strategy of tit-for-tat is more effective in producing cooperation than other more "altruistic" strategies. Furthermore, in the three-choice game this strategy elicited 95% cooperation in the last 30 of the 120 trials.

15. P. Nau - The Investigation of Objects by Rats Placed in a Novel Environment.

An exploring rat spends a great deal of time examining the objects it encounters. Each object is visited repeatedly and contacted with the snout until, after several such visits, the rat will have had experience with its entire surface. Evidence will be presented to show that the location of these contacts, and hence this progression across the object's surface, is highly systematic, depending both on the object's physical characteristics and on the location of previous contacts.

16. C. Barnes - Spatial Memory Deficits in Senescent Rats.

Middle-aged and senescent rats' ability to acquire and retain a task which forced the use of a spatial strategy for its solution was examined. Although both groups improved the accuracy with which they located the goal tunnel, the older rats never attained the same level of proficiency as their young counterparts. Furthermore, when given a one-week retention test, the old rats' performance had deteriorated, while the young rats' performance did not. These results are discussed in terms of previous work which suggests plausible neurophysiological factors underlying the behavioral deficit, and in light of ongoing work which will examine the possibility of producing a "behaviorally senescent" young rat through artificially induced synaptic modification.

17. J. Day & S. MacKay - Lateral S-R Compatibility in a Simple Reaction Time Task as a Function of Handedness and Hand-Writing Posture.

Three groups of subjects, normal right-handers (RN), "normal" left-handers (LN), and "inverted" left-handers who assume a "hooked" hand-writing posture (LI), pressed a key with the index finger of one hand whenever a stimulus appeared to the right or left of a fixation point. Response hand was varied across blocks of trials. For RN and LN subjects, RTs were faster when the response hand was the one ipsilateral rather than contralateral to the stimulus position. However, this ipsilateral hand-field compatibility effect was not observed in LI subjects. Implications for Levy's ipsilateral motor control explanation of LI writing are discussed.

18. J. Fentress - Thoughts on Functional Generalization and Behavioral Development.

The first line of argument is that ethological studies of integrated behavior can benefit by an attempted combined liaison with cognitive psychology and embryology. The second argument is that rules of functional generalization in species-characteristic behavior may provide an important link in this attempted synthesis. Simple actual and potential experiments will be used to illustrate this idea.

learning paradigms?
social learning?
odors
bird song

19. Z. Havkin - Interactive Modes of Wolf (*Canis lupus*) Play.

Several sources of evidence indicate that play behaviour is a well organized phenomenon. Although details of interactions are unique, interactions as whole composites show a high degree of order. In P. Weiss' words: "...individual freedom in the small is compatible with the existence of collective order in the gross...".

20. M. Cynader - Adaptive Modification vs Selective Atrophy in Post-natal Development.

In recent years, many studies have shown that it is possible to alter the recording frequency of cells with particular trigger features by rearing animals in unusual visual environments. To date, it has not been clear whether these changes can be ascribed to the death (or atrophy, or loss of responsiveness) of unstimulated neurons, or whether the trigger features of individual units are altered in a constructive way by unusual exposure. The data to be presented provide strong evidence in favour of adaptive modification of the properties of individual neurons.

21. S. Nakajima - Epinephrine Secretion as an Additional Factor in Producing the Amnesic Effect of Cycloheximide.

My previous study that cycloheximide produces amnesia by suppressing adrenocortical function has been criticized by two groups of investigators. They found that aminoglutethimide, which inhibits adrenocortical function but does not interfere with cerebral protein synthesis, does not produce amnesia. I now report that aminoglutethimide suppresses adrenomedullary function (secretion of epinephrine), and that it produces amnesia if epinephrine is supplemented.

22. N. Allen & J. Barresi - Competence Information and Extrinsic Reward Effects on Intrinsic Motivation.

Extrinsic reward has been shown to decrease rather than increase performance (or liking) of intrinsically motivated activities. It has also been found that performance information (positive or negative) changes intrinsic motivation in the direction of the competence information. In the present study, the interaction of these two variables was investigated. Both factors were found to affect intrinsic motivation in a direction consistent with previous research. However, the magnitude of the reward effect was larger when no competence information was provided. This suggests that subjects attend more to competence information than to extrinsic rewards.

*negative dissonance.
athletics.* *relationship to
researcher's work.*

*drugs which
suppress
adrenal cortex
effect.*

23. B. Rusak - Hypothalamic Lesions Block the Antigonadal Action of Melatonin.

In hamsters an intact pineal gland is necessary for the anti-gonadal effect of exposure to short photoperiods. The pineal product, melatonin, has been shown to produce gonadal regression in short photoperiods and to block gonadal regrowth in long photoperiods; it is presumed that melatonin is a prime pineal antigonadotrophin. Lesions of the suprachiasmatic nuclei (SCN) of the hypothalamus eliminate many circadian rhythms including that of pineal melatonin formation; these same lesions prevent testicular regression in short photoperiods. I have suggested four possibilities for the mechanism of this effect; the simplest of these states that SCN destruction prevents pineal melatonin formation, thereby lowering circulating melatonin levels and permitting the testes to remain functional. On this hypothesis, replacement of melatonin in SCN-lesioned hamsters should exert an antigonadal action. This hypothesis was tested by regressing the testes of hamsters in short photoperiods and using melatonin to block regrowth when returned to a long photoperiod. Melatonin blocked regrowth in intact hamsters but had no effect on SCN-lesioned hamsters. This hypothesis is therefore ruled out, leaving three choices: i) The critical factor is not the level of melatonin but the rhythm of melatonin. ii) The lesions increase gonadotropin levels independently of the pineal. iii) The lesions destroy the site of melatonin uptake so that it cannot influence pituitary function.

24. A. Randich - The Role of the Pituitary-Adrenal Axis in the UCS Pre-exposure Phenomenon.

Experiments tested the hypothesis that the pituitary-adrenal system mediates the decremental effect of prior exposure to an aversive UCS on subsequent acquisition of an excitatory conditioned response, i.e., the UCS pre-exposure phenomenon. Groups of hypophysectomized, adrenal demedullated, adrenalectomized and sham-operated rats received either no pre-exposure treatment or pre-exposure to electric shock, followed by CER conditioning. In no pre-exposure treatment groups, only adrenalectomized rats showed retarded acquisition of the CER. Groups receiving pre-exposure to electric shock generally acquired the CER at a slower rate than their no pre-exposure counterparts. Interpretation of these findings indicated that the pituitary-adrenal system does not mediate the UCS pre-exposure phenomenon obtained with CER conditioning.

Rob
Doug las.

25. F. Baker - Changes in Cytoarchitecture of the Optic Tectum after Reimplantation and Transplantation in the Adult Goldfish as seen by Golgi and Autoradiographic Techniques.

Following reimplantation of tectal tissues or reciprocal transplantation between a tectal tissue and a cerebellar (or forebrain) tissue in the adult goldfish, structural changes in the optic tectum are examined by Golgi staining and autoradiographic methods. In all cases the changes in cytoarchitecture are consistent with the normal pattern of connection of optic fibers from the retina to the optic tectum.

26. M. Ozier - Orthographic and Semantic Access to the Memory Trace.

The episodic memory trace was studied by having subjects attempt to access the memory trace according to different types of retrieval codes. The results indicated that the first letter of the name of a word may function as a retrieval code only if it is intentionally marked as such, whereas the categoric meaning of a word is marked for retrieval purposes automatically. In addition, it was found that the names of words are not linked directly to one another in the lexical dictionary, but the meaning of words have direct connections. These results have implications for theories of encoding specificity, the organization of the mental lexicon, and a speech-production hypothesis of initial-letter cueing effects.

27. K. Bloom - Neonatal Assessment as a Model for Medical/Social Change.

Neonatal assessment is a structured procedure in which defined behaviors are observed and elicited. This process yields not only an objective description of the newborn's medical and behavioral status, but also a generalized affect upon the quality of the observer's subjective judgments. In the former and traditional way, the assessment procedure is used as a dependent variable. In the latter way, assessment is viewed as an independent variable or treatment of the adult (e.g., pediatrician, parent, child-care specialist). The value of structured behavioral assessment for teaching, primary medical care and research will be illustrated with examples from pediatrics, developmental and clinical psychology.

28. T. Whalen - The Power of Learning Theories.

The nature of a theory of learning may be determined using theoretical machines called finite automata. There is some finite automaton which will behave in the same way as any organism. As an organism learns, however, it changes its behaviour. Therefore, there will be a way to change a finite automaton so that it also behaves in this new way. A description of these changes will be equivalent to a theory of learning. If it can be shown that a particular process is required to alter a finite automaton so that it learns appropriate behaviour, a theory of learning must also incorporate this process. For example, if it can be shown that something more than stimulus-response associations are required for a "naive" finite automaton to learn some behaviour, then S-R theories will not be sufficient to account for this learning.

29. J. Barresi & D. Dalzell - The Abstraction of Spatio-Temporal Structure.

A chain of 80 dot patterns was constructed by recursively applying a distortion rule to the pattern constructed from the previous application of the rule. When these patterns are presented visually for 300 msec each in their "natural" order, they appear as a continuously changing shape. Surprisingly, subjects' ability to identify patterns from the sequence is no better when the patterns are presented in their "natural" order than when the sequence of 80 is presented randomly. However, both of these conditions yield much better performance than does a sequence of 80 entirely random patterns. These and other results suggest that resemblance rather than contiguity is the first associative principle.

30. J. Willner - Constraints on Learning: A Re-evaluation.

Although rats preferentially condition to gustatory cues when illness is the reinforcer, the argument that this reflects a specific learning ability selected for by evolution is not especially compelling. Data from experiments on blocking and second-order conditioning which employed nongustatory CS's will be described which suggest that the constraint is on the expression of learning, rather than on the learning itself. The relevance of these findings to general theories of learning will be discussed.

31. J. Raymond - Visual Sensitization with a Large, Flashed Annulus.

The addition of light to the visual system usually acts to desensitize detection mechanisms. In this experiment, specific stimulus conditions are defined that make an exception to this general rule. The results show that the addition of a large, flashed annulus leads to greater sensitivity in detecting a central test spot than when the annulus is not presented. The characteristics of this sensitization effect suggest that different neural mechanisms are involved than those thought to be mediating Westheimer spatial sensitization effects.

32. R. Klein, J. Day, & R. Armitage - Cognitive and Rhythmic Control of Hemisphere Activation.

The two cerebral hemispheres of the human brain are specialized for the performance of different cognitive processes. Metacognition refers to mechanisms that influence which hemisphere is activated and therefore likely to take control of processing in a given situation. Two types of metacognition are distinguished: one is transient and cognitive, the other is rhythmic and automatic. Cognitive metacognition has been demonstrated in split-brain subjects, but its elicitation in normal subjects has been unclear. Rhythmic metacognition has been hypothesized but not clearly demonstrated. We will present clear-cut evidence for both types of metacognition processes in the normal intact brain.

33. N. Bennett - Some Aspects of Communal Pup-Rearing in the Timber Wolf.

Communal pup-rearing is a major co-operative feature of pack social behavior in the timber wolf (Canis lupus) which has not yet been studied in detail. It is known that in the pack, adults assist both mothers and pups in a variety of ways. The present study examined some aspects of pup-rearing in a pack of 12 captive timber wolves and found that assistance to mothers was rendered in a highly selective manner whereas all pups were cared for equally and without distinction. In addition an analysis of the social relationship between pairs of pack members suggests that little aggression occurs during the breeding season between future mothers and their helpers.

34. R. Sutherland - Epithalamic Reinforcement System: Why is the Habenular Complex?

The reinforcing consequences of electrical stimulation at sites in the habenular complex and adjacent structures were investigated in 44 rats. Electrode locations in the stria medullaris, habenular nuclei and fasciculus retroflexus, but not in adjacent thalamic nuclei, support self-stimulation. Lesion studies suggest that connections with the medial raphe and efferent fibers in the stria medullaris generate the rewarding effect of stimulation in the habenular complex.

dominance
relationship
father?